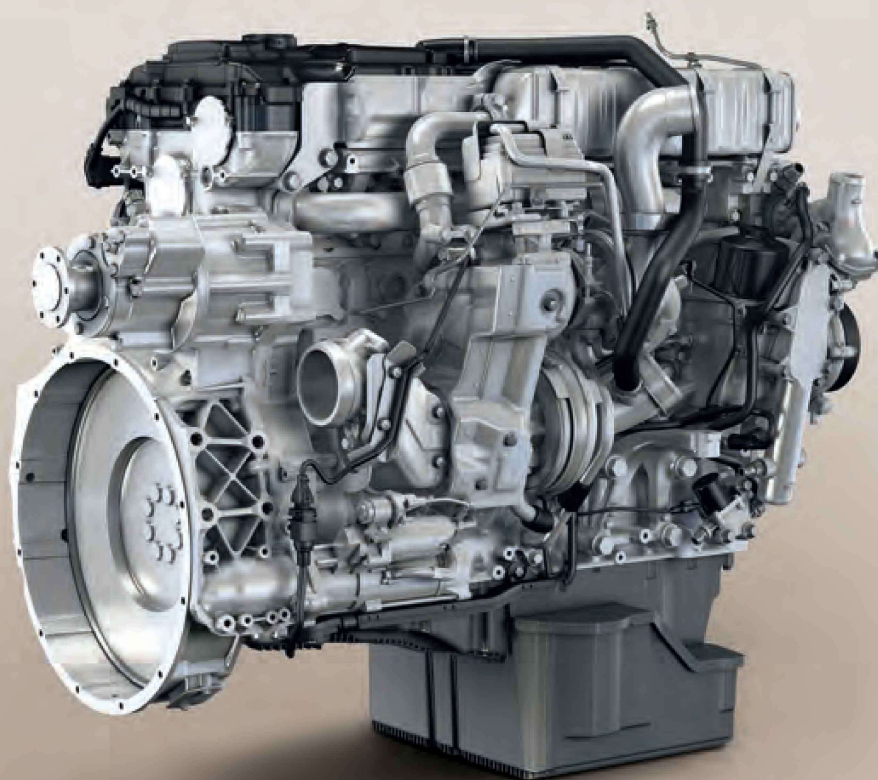




4R/6R 1000

Operating Instructions



Power. Passion. Partnership.

Symbols

WARNING

Warning notes make you aware of dangers which could pose a threat to your health or life, or to the health and life of others.

Environmental note

Environmental notes provide you with information on environmentally aware actions or disposal.

 The purpose of material damage warnings is to draw your attention to risks which could lead to damage to your engine system.

 These symbols indicate useful instructions or further information that could be helpful to you.

► This symbol designates an instruction you must follow.

► Several consecutive symbols indicate an instruction with several steps.

(▷ This symbol tells you where you can find further information on a topic.

▷ ▷ This symbol indicates a warning or an instruction that is continued on the next page.

Welcome

Familiarise yourself with your engine system and read the operating instructions before you use the engine system. This will help you to avoid endangering yourself or others.

The standard equipment and product description of your engine system may vary, depending on individual specifications. This is described on the data card.

The engine systems are constantly updated to be state of the art.

MTU/Mercedes-Benz reserves the right to make changes to the following:

- design
- equipment
- technical features

Descriptions may therefore differ in individual cases from your engine system.

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Engine without exhaust gas after-treatment system

The engines without an exhaust gas aftertreatment system described in the document comply with the "EU Stage IIIA" emissions level.

These are certified in accordance with directive "ECE-R 96 Chapter 5.2.1 for power bands H or I".

The term "EU Stage IIIA" was taken from EU Directive 97/68, is still used in general parlance and is used in this document.

"EU Stage IIIA" emissions level is certified without an associated exhaust gas aftertreatment unit. An exhaust muffler with a silencer is used instead of an exhaust gas aftertreatment unit.

This means that the exhaust muffler used is determined by the vehicle or device and is covered in the operating instructions provided by the vehicle/device manufacturer.

Engine system

In these Operating Instructions, the term "engine system" refers to the combination of the engine and exhaust gas aftertreatment unit. The engine and the emission control system are parts that require certification. The engine system is certified as one unit with the engine and exhaust gas aftertreatment unit assembly with the AdBlue® treatment reactor and the AdBlue® supply system.

Engine systems with exhaust gas aftertreatment can comply with the following emission levels depending on the version:

- EU Stage IV
- US Tier 4
- Japan 2014 - Heisei 26
- Korea Tier 4

Protection of the environment

Environmental note

Daimler AG has a declared policy of comprehensive environmental protection.

The objectives are to use the natural resources which form the basis of our existence on this planet sparingly and in a manner which

takes the requirements of both nature and humanity into account.

You too can help to protect the environment by operating your vehicle in an environmentally responsible manner.

Information and notes on driving in an environmentally responsible and fuel-saving manner can be found in the "Operating notes" section (▷ page 37).

Assembly equipment

These Operating Instructions describe all models and all standard and optional equipment available for your engine system at the time of publication of the Operating Instructions. Country-specific deviations are possible. Note that your engine system may not be fitted with all features described. This also applies to safety-relevant systems and functions. Therefore, the equipment on your engine system may differ from certain descriptions and illustrations.

All of the components in your engine system are listed in the data card of your engine system. Data card (▷ page 60).

Please contact any MTU or MTU-authorized Mercedes-Benz Service Centre if you have any questions about equipment and operation (see addresses in the publication details on the inside of the back cover).

Genuine Mercedes-Benz parts

Environmental note

Daimler AG also supplies reconditioned assemblies and parts which are of the same quality as new parts. For these, the same warranty applies as for new parts.

If you use parts which have not been approved by Mercedes-Benz, the operational safety of the engine system may be jeopardised. This could lead to malfunctions in safety-relevant systems. Use only genuine Mercedes-Benz parts or parts of equal quality. Only use parts that have been approved for your engine type.

Mercedes-Benz checks genuine Mercedes-Benz parts for:

- reliability
- safety
- suitability

Despite ongoing market research, Mercedes-Benz is unable to assess other parts. Mercedes-Benz therefore accepts no responsibility for the use of such parts in Mercedes-Benz vehicles, even if they have been officially approved or independently approved by a testing centre.

In Germany, certain parts are only officially approved for installation or modification if they comply with legal requirements. This also applies to some other countries. All genuine Mercedes-Benz parts meet the approval requirements. The use of non-approved parts may invalidate the vehicle's general operating permit.

This is the case if:

- they result in a change to the vehicle type from that for which the vehicle's general operating permit was granted
- they pose a possible risk for road users
- they adversely affect the emission or noise levels

You can find more information on recommended conversion parts and accessories, as well as permitted technical modifications at any MTU or MTU-authorised Mercedes-Benz Service Centre (▷ page 8).

Always specify the assembly system ID number (AGS ID), the engine number and the number of the exhaust gas aftertreatment unit when ordering genuine Mercedes-Benz parts. You can find the engine number on the identification plate of your engine. You can find the number of the exhaust gas aftertreatment unit on the identification plate of the exhaust gas aftertreatment control module (ACM) (▷ page 60). You can also find the two numbers on the data card (▷ page 60).

The exhaust gas aftertreatment unit and the exhaust gas aftertreatment control unit (ACM) are not used for emissions certificate EU Stage IIIA.

Modifying the engine output

! Increased power could:

- change emission levels
- cause malfunctions
- lead to consequential damage

The operating safety of the engine cannot be guaranteed in all situations.

Any tampering with the engine management system in order to increase the engine power output will lead to a loss of warranty entitlements.

Safety/emergency running program

The engine is equipped with an electronic engine management system that monitors not only the engine and the exhaust gas aftertreatment unit (not for EU Stage IIIA) but also itself (self-diagnosis).

If the electronic control system detects a malfunction, one of the following measures is automatically implemented after an appraisal of the malfunction:

- faults during operation are indicated by the corresponding warning lamp (▷ page 29).
- in conjunction with the electronic engine management system, fault codes with additional information can be shown on a display.
- the system switches to a suitable backup function for the continued, albeit restricted, operation of the engine. This includes torque and engine speed limitation, for example, as well as road speed limitation or constant emergency running speed.

Correct use

The engine system may only be installed as contractually specified.

The manufacturer of the end product is responsible for the correct installation of the engine and the exhaust gas aftertreatment system (not for EU Stage IIIA) in the overall system.

The engine and the exhaust gas aftertreatment system may not be modified. In the event of modification, Mercedes-Benz and MTU do not

accept responsibility for any damage arising as a result.

Correct use of the engine system also requires adherence to the instructions in these Operating Instructions. This also requires compliance with the maintenance intervals and the professional execution of maintenance work. Please observe the Workshop Information System (WIS) (▷ page 8).

Implied warranty

A well-developed network of MTU or MTU-authorized Mercedes-Benz Service Centres is available to carry out maintenance work.

These MTU or MTU-authorized Mercedes-Benz Service Centres:

- have special equipment and tools as well as specialists who receive continuous training
- guarantee that your engine system is repaired and maintained thoroughly and expertly
- carry out all repairs related to implied warranty
- carry out all maintenance work expertly
- confirm in the Maintenance Booklet that the maintenance work has been carried out at the required time
- handle implied warranty claims that are admissible according to the sales contract

Please follow the instructions in this Service Booklet, even if you hand the vehicle/device over to a third party for use or care.

If the specified service/maintenance work is not carried out, the decision as to whether a warranty claim can be accepted or not can be made only after the manufacturer has inspected the claim.

During the implied warranty period, have the prescribed maintenance service for your engine system carried out as follows:

- regularly
- punctually
- at a qualified specialist workshop which has the necessary specialist knowledge and tools to carry out the work required

Confirmation that all service/maintenance work has been carried out at a qualified specialist workshop will be entered in this Maintenance Booklet.

If there are legal requirements on emission control, please note that:

- maintenance on the engines must be carried out according to specific regulations and using special measuring devices
- it is prohibited to modify or tamper with components relevant to emissions

Confirmation that all service/maintenance work has been carried out at a qualified specialist workshop will be entered in this Maintenance Booklet.

Maintenance work does not include repair work. Repair work will be invoiced separately.

You can obtain further information on the maintenance of your engine system from any MTU or MTU-authorized Mercedes-Benz Service Centre.

Stored data

Several of the electronic components in your engine system contain data memories.

These data memories temporarily or permanently store technical information about:

- the engine system state
- events
- malfunctions

In general, this technical information documents the state of a component, a module, a system or the surroundings.

This includes, for example:

- operating conditions of system components, e.g. fluid levels
- the vehicle/device status messages and those of its individual components, e.g. speed, deceleration in movement, accelerator position
- malfunctions and defects in important system components
- the vehicle/device reactions and operating statuses in special driving situations
- ambient conditions, e.g. outside temperature

This data is exclusively technical in nature and can be used to:

- assist in the detection and rectification of faults and defects
- analyse vehicle functions, e.g. after an accident

The data cannot be used to trace the vehicle's movements.

When you use one of the available services, technical information may be read from the event data memory and fault data memory.

Services include, for example:

- repair services
- service processes
- implied warranty and guarantee cases
- quality assurance

The information is read out by employees of the service network (including manufacturers) using special diagnostic testers. Further information is available there if required.

After a fault has been rectified, the information is deleted from the fault memory or is continually overwritten.

Qualified specialist workshop

A qualified specialist workshop has the necessary specialist knowledge, tools and qualifications to carry out the work required on the engine to a professional standard. This is particularly applicable to work relevant to safety.

Observe the notes in the Maintenance Booklet.

Always have the following maintenance work carried out at a qualified specialist workshop:

- work relevant to safety
- service and maintenance work
- repair work
- modifications as well as installations and conversions
- work on electronic components

Please consult an MTU or MTU-authorized Mercedes-Benz partner.

Further applicable documents

These operating instructions describe all models and all standard and optional equipment available for your engine system within the scope of delivery of Daimler AG. The installation of the engine system into the vehicle/device may require additional operating instructions adapted to the vehicle/device and the appropriate use thereof. These additional operating instructions will be provided by the vehicle/

device manufacturer (manufacturer of the end product). The additional operating instructions will describe, in particular, the functions specific to the installation and operation, the use of such functions as well as warning and control mechanisms.

To use the engine correctly, you also require the Maintenance Booklet.

For US-certified off-highway engines you also require the "Emission Warranty" supplement.

Always keep these documents together with the engine, vehicle or device. These documents should be passed on to the new owner if you sell the engine, vehicle or device.

When carrying out maintenance work, you require access to the Workshop Information System (WIS) via the Internet. This access is subject to a fee.

Current information on the system and prices can be found at this web address: <http://service-parts.mercedes-benz.com>. Click on "EPC, WIS/ASRA" in the "Service and parts information" tab and then on "WIS".

You can log in by clicking on "Register" on the right-hand side.